

Varakumar G

Security Developer | Detection Engineer | Vulnerability Research | ICS/OT Security
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GitHub | GitLab | Portfolio | Letter of Recommendation

PROFESSIONAL SUMMARY

Security Developer / Detection Engineer with 2 years of experience developing vulnerability detection logic, security automation, and cloud compliance checks for enterprise vulnerability management platforms. Authored 3,200+ NASL/NSE detection plugins covering ICS/SCADA protocols including Modbus, BACnet, EtherNet/IP, DNP3, PROFINET, and SNMP, expanding enterprise detection coverage by 700%. Experienced with OpenVAS, Nessus, Qualys, CloudSploit, Python, Bash, Node.js, and AWS, Azure, GCP security. Active open-source contributor maintaining Android KernelSU, NetHunter, and Linux kernel projects while building offensive security tooling and AI-driven automation. Seeking opportunities in Detection Engineering, Vulnerability Research, Product Security, Security Engineering, Offensive Security, Cloud Security, ICS Security and OT Security.

KEY ACHIEVEMENTS

- Authored custom NASL scripts to scale **SCADA CVE detection coverage 8×** ($\approx 400 \rightarrow 3,200+$ ICS/SCADA vulnerabilities) in under 12 months using OpenVAS and Nessus, growing the company's SCADA CVE database by 700%.
- Sole owner of **CloudSploit** across AWS, Azure, and GCP: maintained CIS benchmark compliance, developed custom Node.js detection plugins, and resolved feed conflicts in distributed scanner environments.
- Wrote detection logic for **ICS/SCADA protocols** (Modbus, BACnet, EtherNet/IP, SNMP), closing OT vulnerability gaps across enterprise scanning platforms.
- Built a **multi-agent AI dev pipeline** where an orchestrator proposes each change and separate reviewer and tester agents must both approve before anything ships; every fix is written back into the agent rules to cut repeat errors.

TECHNICAL SKILLS

Vulnerability Management: OpenVAS, Nessus, Qualys, CloudSploit

Offensive Skills: Penetration Testing, Vulnerability Assessment, Exploitation, OSINT, Social Engineering, Active Directory

Offensive Tools: Burp Suite, Nmap, Aircrack-ng, Bettercap, Kismet, Wireshark, Reaver, Metasploit

Detection & Scripting: NASL, NSE (Nmap), Node.js Plugins, Bash / iptables, Python Automation

ICS/SCADA Protocols: SNMP, Modbus, BACnet, EtherNet/IP, DNP3, PROFINET

Cloud Platforms: AWS, Microsoft Azure, Google Cloud Platform (GCP)

Frameworks / Standards: MITRE ATT&CK, OWASP Top 10, NIST, CIS Benchmarks, NVD

Languages / Databases: Python, Node.js, Bash, NASL, NSE — MongoDB, MySQL

OS / Kernel / Mobile: Arch Linux, Kali Linux, Android AOSP, KernelSU, SUSFS, NetHunter, WireGuard

PROFESSIONAL EXPERIENCE

Holm Security AB | Security Developer

Remote (Sweden) | Apr 2023 – Mar 2025

- Hand-authored NASL scripts that grew SCADA CVE coverage from ≈ 400 to 3,200+ vulnerabilities in 12 months using OpenVAS and Nessus, a 700% expansion of the company's SCADA CVE database.
- Sole owner of **CloudSploit** across AWS, Azure, and GCP: maintained CIS benchmark compliance, shipped custom Node.js detection plugins, and resolved upstream feed conflicts in distributed environments.
- Wrote NASL detection scripts for **ICS/SCADA protocols** (Modbus, BACnet, EtherNet/IP, DNP3, PROFINET, SNMP), closing OT vulnerability gaps in enterprise scanning coverage.
- Independently managed the full plugin lifecycle (research, NASL authoring, QA, deployment) and conducted structured assessments across Azure, GCP, and AWS.

Independent Research & Open Source | Security Researcher

Self-Directed | 2024 – Present

- **Android Kernel / NetHunter, OnePlus 9/9 Pro:** Ported KernelSU Next and SUSFS for root management and filesystem virtualisation. Built and maintain NetHunter support: Wi-Fi monitor mode and packet injection (qcacl-d-3.0), USB HID emulation, and tool support for Aircrack-ng, Bettercap, Kismet, Wireshark, and Reaver. Ships as a stable, battery-efficient kernel across major AOSP ROMs with active community support.
- **Agentic AI Dev Pipeline:** Built a multi-agent system where an orchestrator proposes changes and separate reviewer and tester agents must both approve before anything ships. Each fix is written back into the agent rules, reducing repeat errors over time. Applied offensive security knowledge to identify and fix security flaws flagged during the automated review cycle.

PROJECTS

- **WireGuard VPN: Self-Hosted Server** | Bash & Linux Networking | Aug 2025
Set up a WireGuard VPN on Ubuntu with automated peer provisioning, routing, and iptables rules via Bash to isolate and manage VPN traffic.
- **Bill Manager: Self-Hosted Billing & Inventory PWA** | React, Node.js, Prisma, PostgreSQL, Docker | 2026
Full-stack billing and batch-inventory system with FEFO stock tracking, AI bill scanning (extracts items, batch numbers, expiry dates), a PDF invoice designer, and email dispatch. Packaged as a single Docker image via GitHub Actions to Docker Hub with Caddy auto-TLS for local-LAN and VPS installs.

CERTIFICATIONS

- Offensive Security Certified Professional (OSCP) | OffSec | *In Progress*
- TCM Security: Practical Ethical Hacking • Windows & Linux Privilege Escalation • OSINT Fundamentals • External Pentest Playbook

EDUCATION

Malla Reddy College of Engineering & Technology, Hyderabad, India
Bachelor of Technology (B.Tech), Computer Science

2018 – 2022